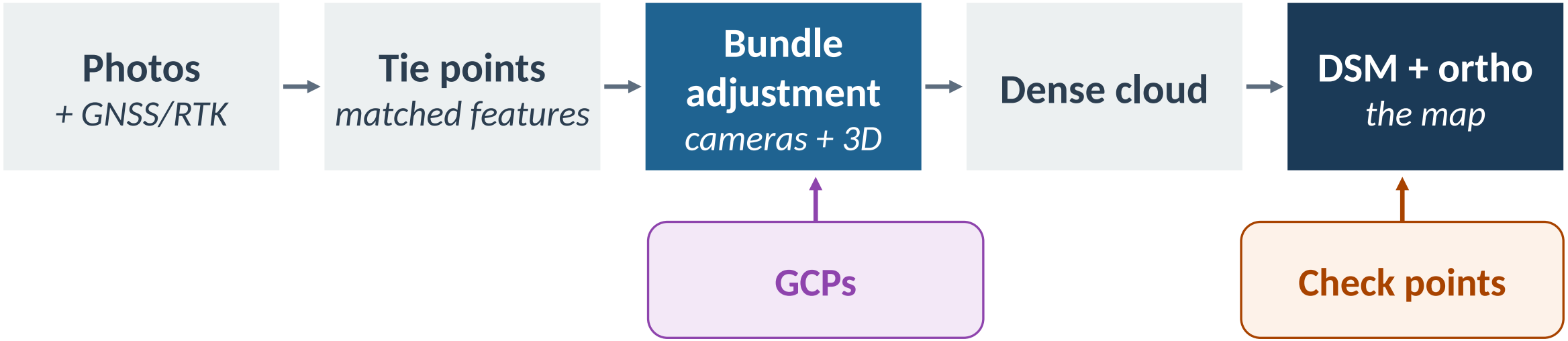
Drone mapping

Ground control, photos for structure from motion, mission planning



From photos to a map



- **Today:** the input – control points, photos, flight plan
- **After the flights:** WebODM, orthomosaic, DSM, accuracy

Wikipedia: Photogrammetry · Structure from motion · Bundle adjustment · Orthophoto · Digital elevation model

Today

- **Control:** ground control points and check points, targets
- **Photos:** overlap, pixel size, sharpness, light
- **Plan:** from flying height to flight lines and flight time
- **Outside:** measure the control points with RTK



1 Control

Ground control points and check points

Photos → map | **Control** | Photos | Plan



Ground control and check points

Ground control point (GCP)

- Known coordinates, visible in the photos
- **Used** in the bundle adjustment
- Fixes position, scale, rotation, height

Check point

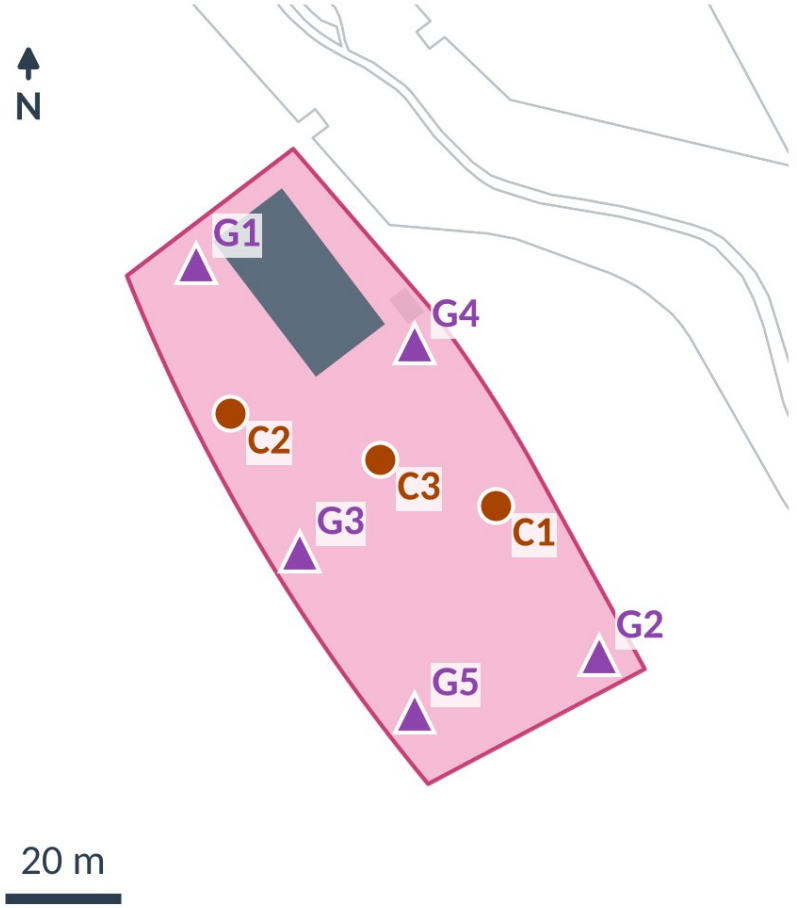
- Measured the same way
- **Not used** in the processing
- Tests the result: error in E, N, H

Pause: why can a GCP not tell you how good your map is?

Wikipedia: Ground control point · Accuracy and precision · Root mean square deviation



How many, and where?



- **≥ 5 GCPs**: edges and middle
- Each one in ≥ 5 photos
- **Check points** in between, never used as GCPs
- Open sky, flat ground, away from walls
- Final places: decided in the field

Wikipedia: Ground control point · Bundle adjustment

Map: ALKIS (simplified WFS) and height benchmarks: Geobasis NRW, dl-de/zero-2-0; layout: R. Becker; rules: OpenDroneMap documentation (GCP)

The drone has RTK: do we still need GCPs?

camera positions from	accuracy	GCPs	check points
GNSS alone	metres	needed	needed
DGNSS (code)	dm … 1 m	needed	needed
RTK / PPK on the drone	1–2 cm	optional	needed

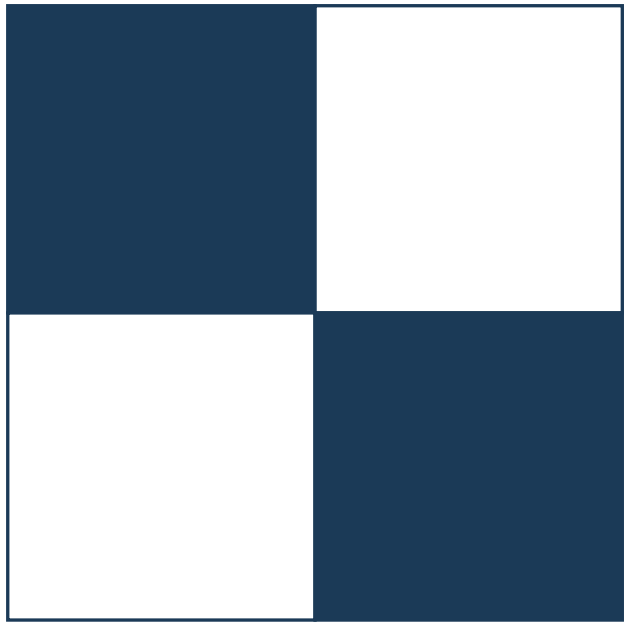
- **No SAPOS on site?** own base + LoRa (relative cm), or PPK afterwards
- **DGNSS** is not enough for control: a GCP must beat one pixel

Wikipedia: Real-time kinematic positioning · Differential GNSS

Targets you can find in the photos

50 cm panel	RGB	MS
at 25 m	73 px	44 px
at 60 m	31 px	18 px

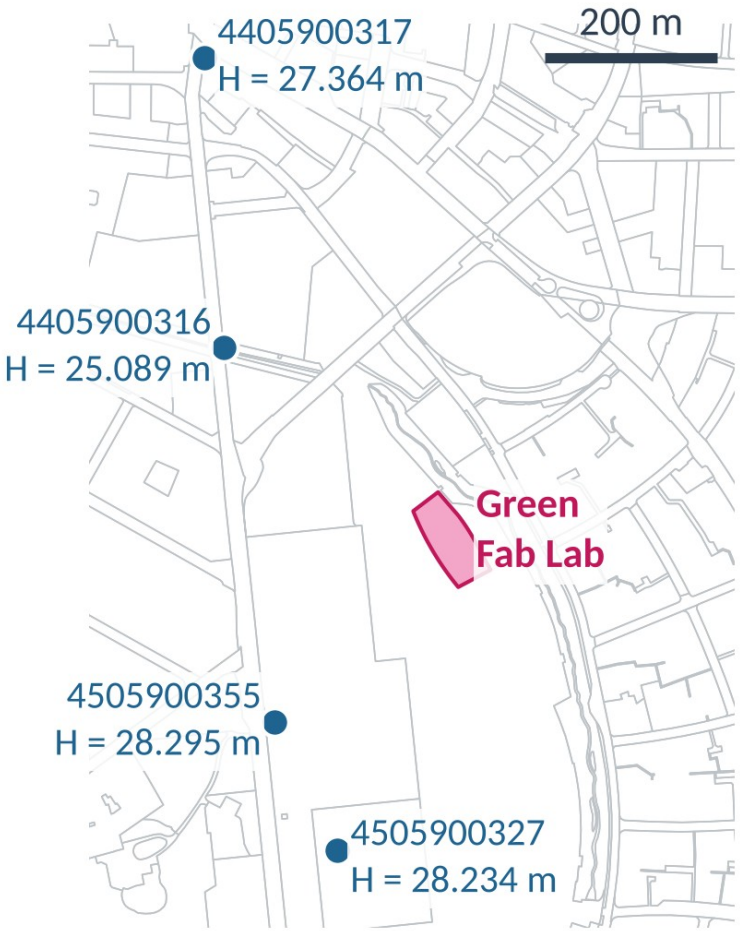
- Measure the centre: FIX, ≈ 30 s
- Measure again later
- Photo of each panel + name
- Leave it until the drone has landed



matte, flat, weighted

Wikipedia: Fiducial marker · Ground sample distance

Survey marks near the Green Fab Lab?



- **Position marks:** all destroyed
- **Height benchmarks:** bolts in walls, levelled 2022
- No pole on a wall bolt
- **→ coordinates from SAPOS RTK, measured twice**

Height benchmarks (HFP CSV) and land use (ALKIS): Geobasis NRW, dl-de/zero-2-0; status: TIM-online / AFIS WMS

GCPs for WebODM

```
EPSG:25832
329851.42 5708012.37 26.91 2734 1880 DJI_0123.JPG G1
329851.42 5708012.37 26.91 1022 2410 DJI_0124.JPG G1
329893.10 5707990.85 27.05 3911  760 DJI_0157.JPG G2
```

- **Per line:** E N H, pixel x y, photo, name
- Pixels: click in the WebODM GCP interface
- **One height system:** NHN vs. ellipsoid ≈ 45 m

Format: OpenDroneMap documentation, Ground Control Points (docs.opendronemap.org/gcp)



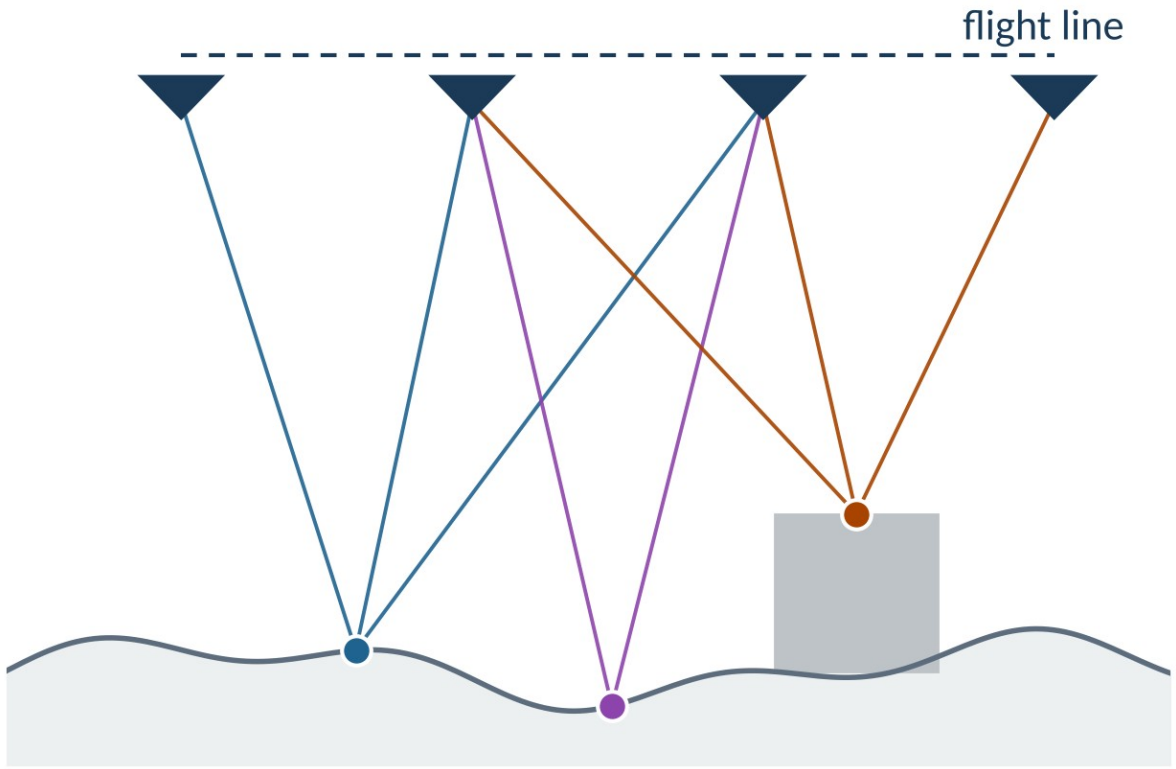
2 Photos

What structure from motion needs

Photos → map | Control | **Photos** | Plan



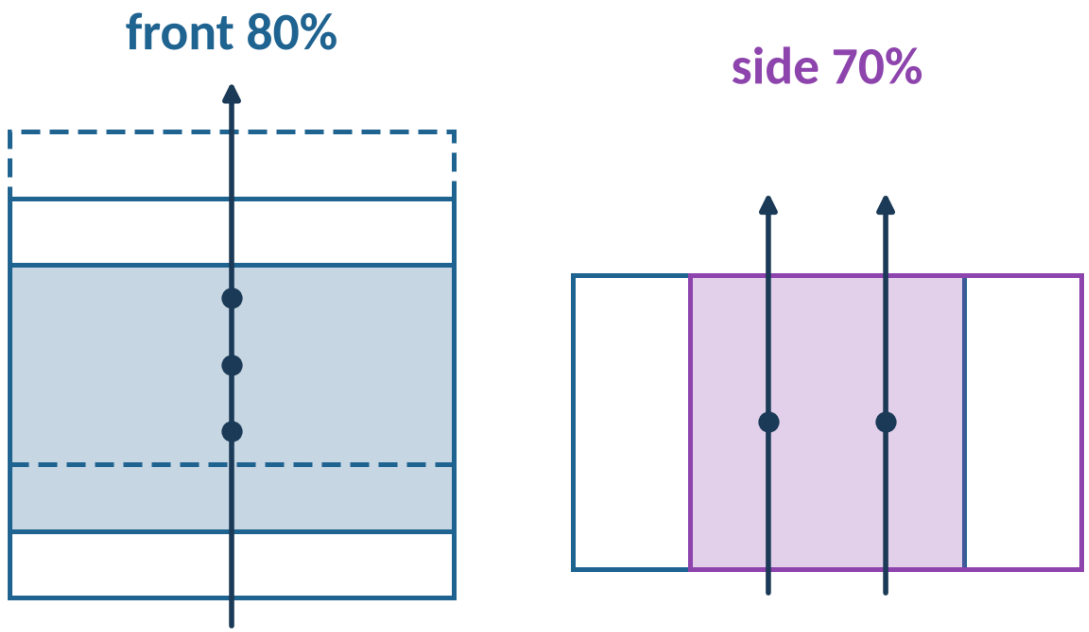
Structure from motion



- Same feature in ≥ 3 photos (better 5+)
- Needs texture
- Scene must stand still
- Walls only from oblique photos

Wikipedia: Structure from motion · Feature (computer vision) · Bundle adjustment

Overlap



arrows: flight lines, dots: photo positions

- **Maps (2D):** 70–80 %
- **Trees, buildings:** ≈ 80–83 %
- **3D:** + cross grid, camera at 45°
- Front overlap is cheap, side overlap costs flight time

Wikipedia: Photogrammetry · Image stitching

Numbers: OpenDroneMap documentation, Flying tips (docs.opendronemap.org/flying)

Pixel size on the ground (GSD)

GSD = H × (width of the view on the ground per metre) / pixels

height	RGB pixel	MS pixel	RGB photo
25 m	0.7 cm	1.1 cm	36 × 27 m
40 m	1.1 cm	1.8 cm	58 × 43 m
60 m	1.6 cm	2.7 cm	86 × 65 m
120 m	3.3 cm	5.5 cm	173 × 130 m

Mavic 3M, GSD in cm: RGB ≈ H / 36.6, multispectral ≈ H / 21.9 (H in m)

Wikipedia: Ground sample distance

Camera: DJI Mavic 3M specifications (ag.dji.com/mavic-3-m/specs)



Sharp photos

blur = speed × exposure time (keep < ½ pixel)

speed	exposure	blur	at 25 m (RGB 0.68 cm)
2.2 m/s	1/1000 s	2.2 mm	0.3 px
7.7 m/s	1/1000 s	7.7 mm	1.1 px
7.7 m/s	1/250 s	30.8 mm	4.5 px

- Low light → longer exposure → fly slower
- **RGB: mechanical shutter** · MS: electronic

Wikipedia: Motion blur · Shutter speed · Rolling shutter



Light, weather, scene

- **Even light:** overcast is fine, broken clouds are not
- **November sun:** ≈ 23° at noon → shadows 2.4 × object height
- **Wind:** moving trees, blurred photos
- **Multispectral:** light sensor + reflectance panel before and after
- **Scene:** no water, glass, moving people or cars

Wikipedia: Solar zenith angle · Multispectral imaging

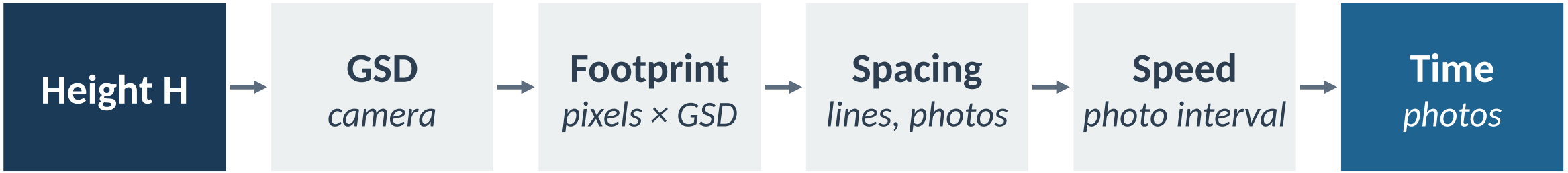
3 Plan

From flying height to flight lines

Photos → map | Control | Photos | **Plan**



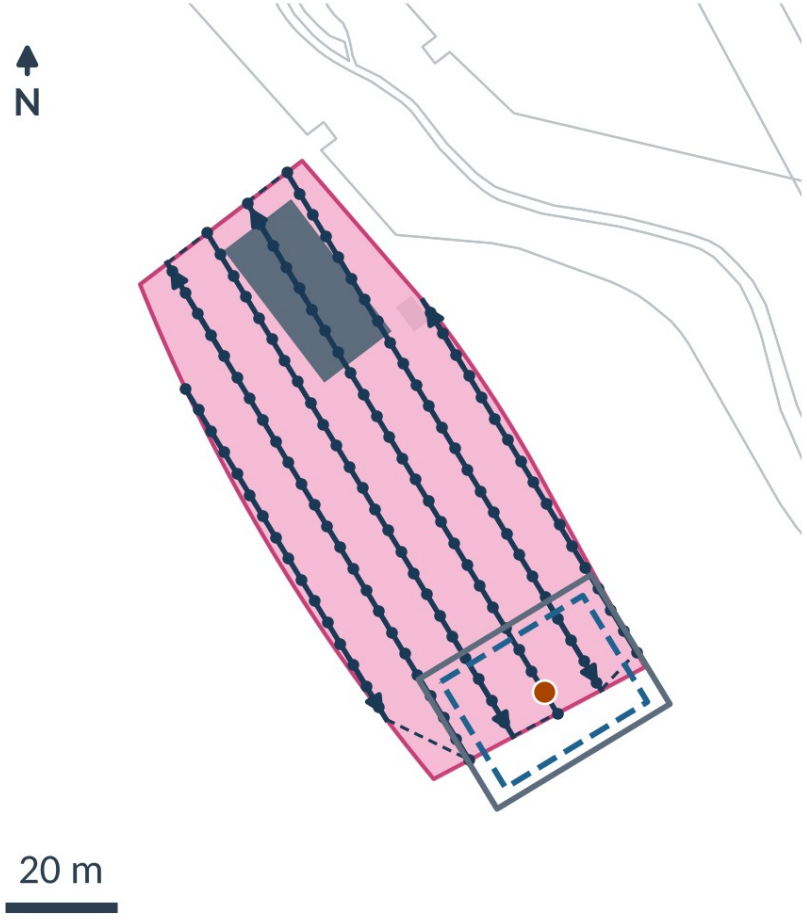
From flying height to flight time



- **Line spacing** = footprint across × (1 − side overlap)
- **Photo spacing** = footprint along × (1 − front overlap)
- **Speed** = photo spacing / photo interval (RGB 0.7 s, MS 2 s)
- **Lower**: finer pixels, more lines, longer flight

Wikipedia: Ground sample distance · Photogrammetry

The Green Fab Lab plan



all cameras, 25 m	
pixel RGB / MS	0.7 / 1.1 cm
lines, spacing	6 × 8.9 m
photo every	4.4 m
speed	2.2 m/s
photo positions	131
flight time	≈ 5 min

Map: ALKIS (simplified WFS) and height benchmarks: Geobasis NRW, dl-de/zero-2-0; plan: R. Becker (common.mission); camera: DJI Mavic 3M specs

All cameras: the multispectral camera decides

at 25 m	RGB	multispectral
pixel	0.68 cm	1.14 cm
photo on the ground	36 × 27 m	30 × 22 m
line spacing (70 %)	10.8 m	8.9 m
photo spacing (80 %)	5.4 m	4.4 m
photo interval	0.7 s	2 s
max. speed	7.7 m/s	2.2 m/s

Pause: what changes if we fly at 40 m?



Before take-off

Rules

Registered operator, certified pilot

Visual line of sight, ≤ 120 m

Not over uninvolved people

Geo-zones: dipul.de

On site

Public park: watch the paths

Wind, rain, light

Batteries, home point

RTK fix before take-off

Wikipedia: Regulation of unmanned aerial vehicles

Rules: Regulation (EU) 2019/947 (open category); geo-zones: dipul.de. The pilot in command decides.



Plan in Python, fly with DJI Pilot 2



- You compute the numbers; the app flies the lines
- Same height and overlap in both
- **Homework:** your plan for the Green Fab Lab

Wikipedia: Keyhole Markup Language

Summary

- GCPs fix the model, check points test it – never mix them
- Drone RTK: GCPs optional, check points still needed
- Every point in ≥ 3–5 photos: 70–80 % overlap, texture, a still scene
- Height sets pixel size, lines, photos and time
- **All cameras: the multispectral camera decides**



Sources and further reading (1/2)

From photos to a map: Photogrammetry · Structure from motion · Bundle adjustment · Orthophoto · Digital elevation model

Ground control and check points: Ground control point · Accuracy and precision · Root mean square deviation

How many, and where?: Ground control point · Bundle adjustment

The drone has RTK: do we still need GCPs?: Real-time kinematic positioning · Differential GNSS

Targets you can find in the photos: Fiducial marker · Ground sample distance

Structure from motion: Structure from motion · Feature (computer vision) · Bundle adjustment

Overlap: Photogrammetry · Image stitching

Pixel size on the ground (GSD): Ground sample distance

Sharp photos: Motion blur · Shutter speed · Rolling shutter

Light, weather, scene: Solar zenith angle · Multispectral imaging

From flying height to flight time: Ground sample distance · Photogrammetry

Before take-off: Regulation of unmanned aerial vehicles



Sources and further reading (2/2)

Plan in Python, fly with DJI Pilot 2: Keyhole Markup Language

Text distilled from Wikipedia (CC BY-SA 4.0), the OpenDroneMap documentation (docs.opendronemap.org), the DJI Mavic 3M specifications and the ASPRS Positional Accuracy Standards, Edition 2 (2023). Maps: ALKIS and height benchmarks, Geobasis NRW (dl-de/zero-2-0). Figures: R. Becker (make_slides.py), CC BY-SA 4.0.



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